

Physikalisches Grundpraktikum

Versuchsprotokoll: Pendel

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Gruppe B1

Measuring Gravitational Acceleration Using a Pendulum

Aufgabe 1 Determining the Pendulum Length

Describe your procedure for determining the length of the pendulum. State your results for the pendulum length l_p and the diameter d_p of the pendulum bob, along with the corresponding measurement uncertainties.

To determine the length of the pendulum and the diameter of the pendulum bob, we conducted measurements independent of the person performing them and subsequently arrived at identical results. We divided our measuring apparatus into three sections so that we could measure them more easily and then add the results of the sections together. For illustration purposes, a sketch is attached (1). First, we measured the length of the pendulum, from the upper edge of the U-shaped suspension to the outer edge of the pendulum bob; this length is referred to as l_1 below. We performed the measurement using a tape measure with a minimum scale of 1 mm and a tolerance of ± 0.7 mm. We both obtained a length of 615 mm. To calculate the uncertainty for this part, we must determine σ_{reading1} and σ_{calib1} . The general formula for this is:

$$\sigma = \frac{\Delta}{\sqrt{12}} \quad (1)$$

Where Δ denotes the width of the uncertainty interval. For our values σ_{reading1} and σ_{calib1} , the following values result:

$$\sigma_{\text{reading1}} = \frac{1}{\sqrt{12}} = 0.29 \text{ mm}, \sigma_{\text{calib1}} = \frac{1.4}{\sqrt{12}} = 0.40 \text{ mm}. \quad (2)$$

Thus, the total error on l_1 is:

$$\sigma_{l1} = \sqrt{\sigma_{\text{reading1}}^2 + \sigma_{\text{calib1}}^2} = \sqrt{0.29^2 + 0.40^2} = 0.494 \text{ mm} \quad (3)$$

And for l_1 , the overall result is then:

$$l_1 = 615 \text{ mm} \pm 0.494 \text{ mm} \quad (4)$$

The measurement was taken with the pendulum rod suspended so that it could not bend. Since the pendulum rod does not align exactly with the pendulum's pivot point, but instead hangs from the axis of rotation in a U-shaped bracket. We must now also calculate this section. For the measurement, we chose to use a vernier caliper to precisely determine the length from the tip of the needle to the upper edge of the underside of the U-shaped bracket. After taking a single measurement, we both arrived at a result of 27.15 mm. The vernier scale has a smallest scale unit of 0.05 mm; we use this again to calculate σ_{Reading2} and σ_{Calib2} , just as we did for l_1 , using the formula from 1. The total error is again given by:

$$\sigma_{l2} = \sqrt{\sigma_{\text{reading2}}^2 + \sigma_{\text{calib2}}^2} = \sqrt{\frac{0.05^2}{\sqrt{12}} + \frac{0.1^2}{\sqrt{12}}} = \sqrt{0.0144^2 + 0.0289^2} = 0.0323 \text{ mm} \quad (5)$$

Therefore, for l_2 , we have:

$$l_2 = 27.15 \text{ mm} \pm 0.0323 \text{ mm} \quad (6)$$

The only thing missing now is the diameter of the pendulum mass itself, denoted below as d_p . For d_p , we used the caliper again, and we both arrived at the same result of 80 mm. We have already calculated the errors associated with the vernier caliper in 5, so we arrive at a result for d_p :

$$d_p = 80 \text{ mm} \pm 0.0323 \text{ mm} \quad (7)$$

To find the total length l_p , we now need the following formula:

$$l_p = l_1 + l_2 + \frac{d_p}{2} \quad (8)$$

And substituting the values, we get:

$$l_p = 615 \text{ mm} + 27.15 \text{ mm} + \frac{80 \text{ mm}}{2} = 682.15 \text{ mm} \quad (9)$$

For the total uncertainty σ_{lp} , we obtain the following values:

$$\sigma_{lp} = \sqrt{\sigma_{l1}^2 + \sigma_{l2}^2 + \left(\frac{\sigma_{dp}}{2}\right)^2} = 0.495 \text{ mm} \quad (10)$$

The following sketch illustrates once again which lengths (l_p , l_1 , l_2 , and d_p) cover which sections in the experiment.

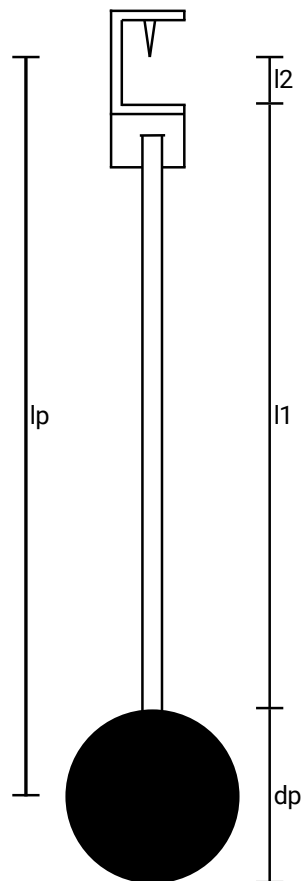


Figure 1: Schematic representation of the pendulum geometry with the following dimensions: l_2 (distance from the pivot point of the angle sensor to the upper edge of the underside of the U-shaped suspension, vernier caliper), l_1 (length of the rod to the pendulum body, measuring tape), l_p (total length from the pivot point to the center of mass) and d_p (diameter of the pendulum body, vernier caliper).

Aufgabe 2 Determining the Period Length

Determine the period and the measurement uncertainty for your series of measurements by performing a suitable model fit. As an example, plot one of the oscillatory processes measured with the pendulum. Show appropriate sections, if necessary, so that the individual oscillations can be clearly identified. Plot the fitted model, create the residual plot, and specify the goodness of fit. Discuss your results. Tabulate the period durations from your individual measurement series and state your final result for the period duration.

The deflection of the pendulum is converted by the Hall sensor into a voltage U . This voltage is time-dependent and oscillates in a damped manner around the baseline voltage U_b :

$$U(t) = U_0 \cdot e^{-\delta \cdot t} \cdot \cos(\omega \cdot t + \phi) + U_b \quad (11)$$

with amplitude U_0 , damping constant δ , angular frequency ω , phase φ , and base voltage U_b . To determine the statistical measurement uncertainty σ_U of the Hall probe, a noise measurement was performed without the pendulum attached. Since there is no oscillation in this case, this measurement does not provide a value for the amplitude U_0 , but rather for the baseline voltage U_b of the sensor, as well as its standard deviation. To obtain a more accurate value, the noise measurement was repeated five times, and for each measurement, the mean and standard deviation were calculated.

U_b	σ_U
−0.0945 V	0.0022 V
−0.1036 V	0.0022 V
−0.1068 V	0.0026 V
−0.1062 V	0.0022 V
−0.1071 V	0.0023 V

Table 1: Noise Measurements on the Hall Probe

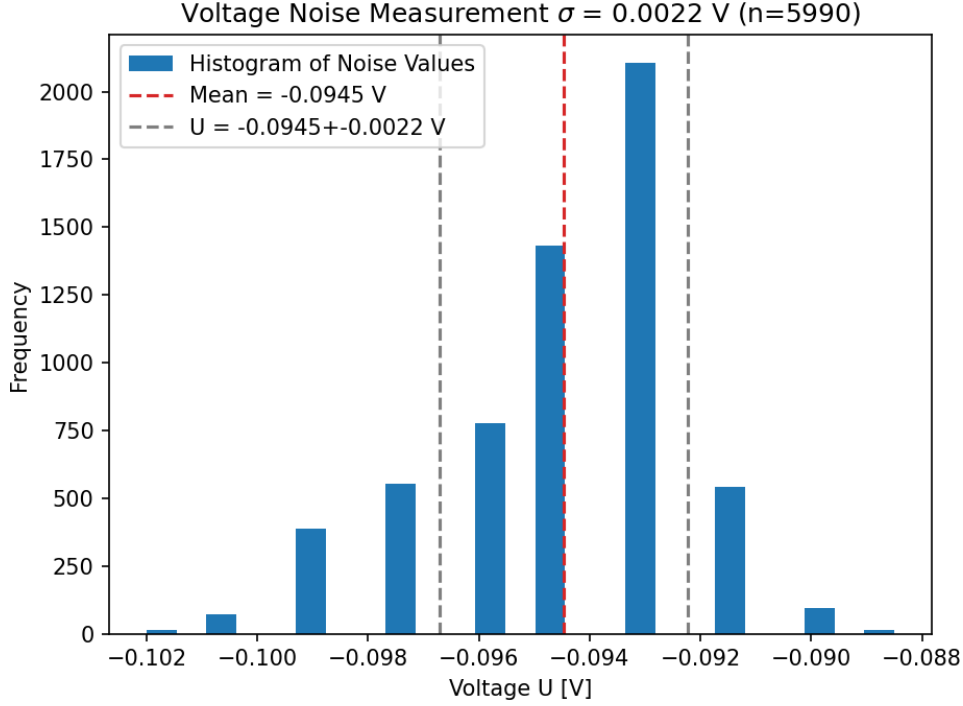


Figure 2: Histogram of the first noise measurement (excluding the first 10 recorded values)

To calculate the results, we filtered out the first ten values from each recording, as these values clearly did not match the rest of the data. We attribute this to the fact that the Hall probe first had to adjust to the measurement, as an internal capacitance may have needed to be charged. As a result, the mean of the five individual measurements for U_b and the mean of the five standard deviations are used as the measurement uncertainty σ_U :

$$U_b = -0.1036 \text{ V} \pm 0.0023 \text{ V} \quad (12)$$

This uncertainty σ_U is used as a weight in the model fitting according to Equation 11. All five parameters of the fit (U_0 , δ , ω , φ , and U_b) are determined freely by the fit. The value for U_b obtained from the noise measurement therefore does not serve as a fixed input value, but rather as an independent plausibility check for the value determined by the fit.

To determine the period T of the system, we'll use a trick. First, we determine the period T of the pendulum without the mass. Then, we move the mass along the pendulum rod until the period of the pendulum with the mass exactly matches the period T we previously determined for the pendulum without the mass. To avoid having to fit a curve to the series of measurements, we initially determined T using the following equation:

$$T = \frac{t_{\text{end}} - t_{\text{begin}}}{n} \quad (13)$$

where t_{start} and t_{end} are the start and end of a period, respectively. n is the

number of periods that have elapsed in the interval from t_{start} to t_{end} . We then obtain the following values for T so that we can later compare these periods with the periods when a mass is suspended from the pendulum.

T_1	1.6582 s
T_2	1.6582 s
T_3	1.6582 s

Table 2: Measurement of the Period of a Pendulum Without a Mass

We then attached the mass to the pendulum and adjusted it upward until, using Equation 13, we obtained the same period as we had without the mass. We then performed three additional control measurements, which yielded the same values as those in Table 2.

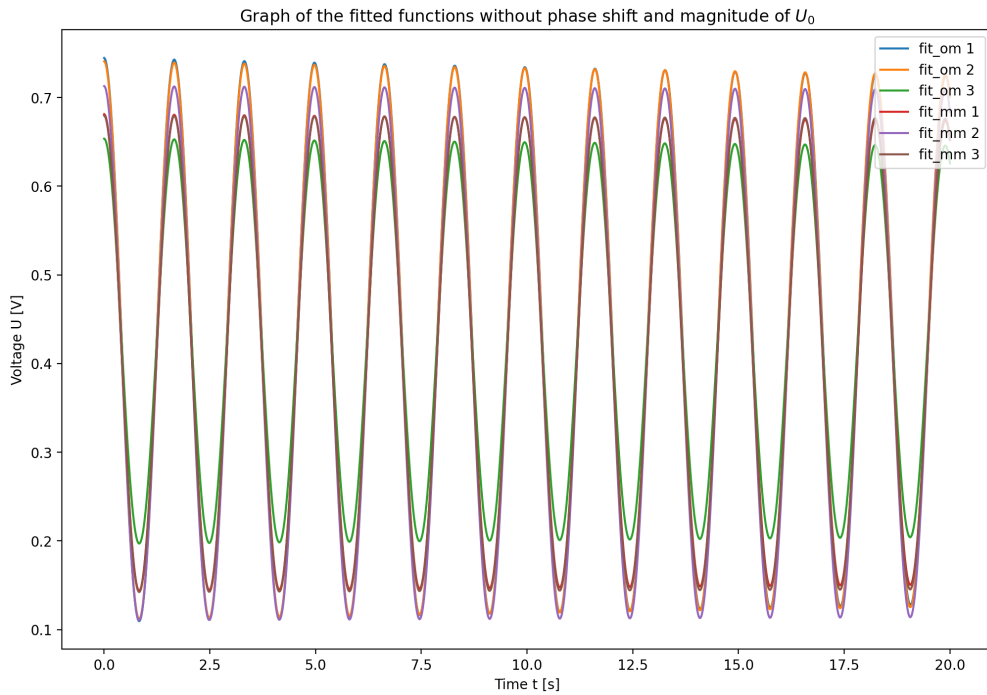


Figure 3: Comparison of the periods with a phase shift set to $\phi_0 = 0$ and the magnitude of $U_0 \rightarrow |U_0|$

Now that we have found the desired position of the mass, we have moved on to the actual main measurements, each with a measurement duration of 180 s per measurement series, in order to obtain well over 100 complete periods. A total of ten measurement series were recorded in order to determine the uncertainty in T , and thus in g , as accurately as possible.

For each of the ten measurement series, a fit of the function 11 to the measurement data was performed, weighted by the uncertainty σ_U from the noise measurement. Figure 4 shows, as an example, measurement series 1 with the fitted model, as well as the corresponding residual plot, each for a segment lasting a few seconds, in which individual oscillations are clearly visible.

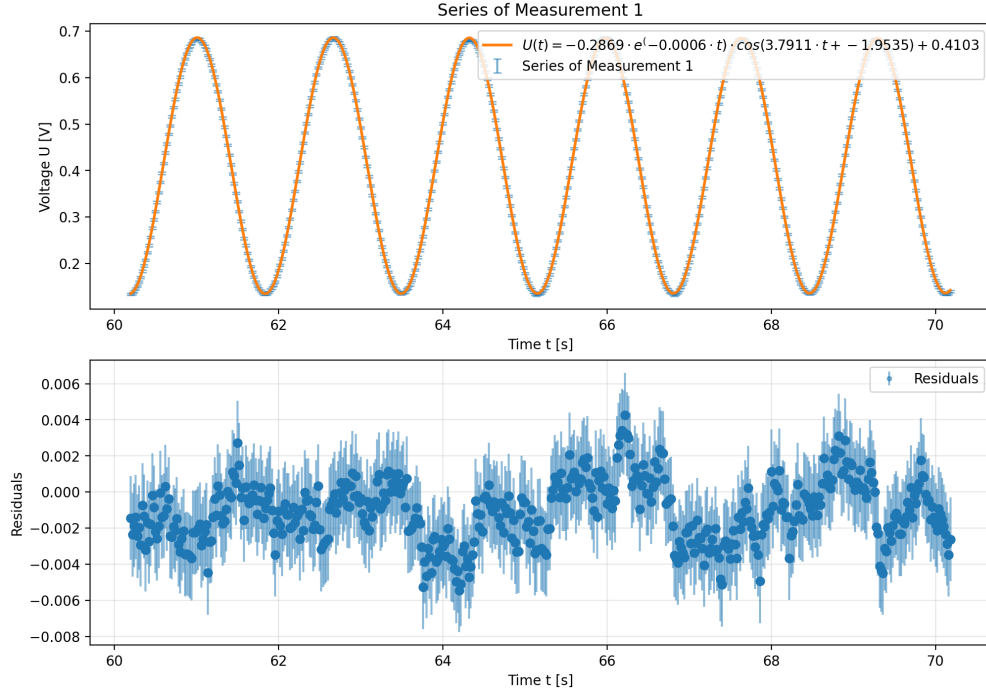


Figure 4: Top: Measurement series 1 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

The residuals are scattered around zero, with no discernible systematic pattern (no trend, no periodicity in the residuals themselves), which suggests that the damped cosine model accurately describes the measurement data. The reduced χ^2 serves as a quantitative measure of the quality of the fit:

$$\frac{\chi^2}{\text{dof}} = \frac{1}{n - p} \sum_j \left(\frac{U_j - U_{\text{fit}}(t_j)}{\sigma_U} \right)^2 \quad (14)$$

with $p = 5$ free parameters. For measurement series 1, we obtain $\chi^2/\text{dof} = 1.0095$; averaged over all 10 measurement series, $\overline{\chi^2/\text{dof}} = 0.8999$.

The fits of the ten main measurement series yield the period durations T_i listed in Table 3, along with their respective fit uncertainties $\sigma_{T,i}$.

Measurement	T [s]	σ_T [s]
1	1,6573	$1,0770 \times 10^{-6}$
2	1,6575	$0,9937 \times 10^{-6}$
3	1,6573	$1,1030 \times 10^{-6}$
4	1,6572	$1,2074 \times 10^{-6}$
5	1,6570	$1,3523 \times 10^{-6}$
6	1,6569	$1,5228 \times 10^{-6}$
7	1,6568	$1,6939 \times 10^{-6}$
8	1,6567	$1,8517 \times 10^{-6}$
9	1,6567	$2,0373 \times 10^{-6}$
10	1,6571	$1,2712 \times 10^{-6}$

Table 3: Period T_i and fit uncertainty $\sigma_{T,i}$ of the ten main measurement series.

A slight trend in T_i is noticeable from the fourth measurement series through the ninth measurement series (falling from 1.6572 s to 1.6567 s), before measurement series 10 returns closer to the initial value. This suggests a slow systematic drift during the measurement series (e.g., slight warming of the system, minimal readjustment), which is not captured by the pure sensor noise σ_U .

To determine the mean period length $\bar{T}$ and its uncertainty σ_T , we directly use the values T_i obtained from the model fits:

$$\bar{T} = \frac{1}{n} \sum_{i=1}^n T_i \quad (15)$$

$$\sigma_T = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (T_i - \bar{T})^2} \quad (16)$$

The final result is:

$$\bar{T} = 1.6571 \text{ s}, \quad \sigma_T = 2.76 \cdot 10^{-4} \text{ s} \quad (17)$$

This variation between the measurement series is significantly greater than the fit uncertainty $\sigma_{T,i}$ of the individual fits (on the order of 10^{-6} s, see Table 3). This means that it is not the sensor noise σ_U , but rather the drift between the measurement series described above that is the dominant source of error for T . Both components are included separately in the error calculation for g in the next section as σ_{stat} (variation between measurement runs) and σ_{fit} (mean fit uncertainty).

Aufgabe 3 Determining the Acceleration Due to Gravity

Calculate the acceleration due to gravity. Perform an error analysis to determine the measurement uncertainty. State the individual contributions that make up the total error as relative errors. Discuss which error dominates the total error. Compare your final result with the relevant value reported in the literature.

From the angular frequency ω of the damped oscillation (Equation 11) and the

pendulum length l_p , the acceleration due to gravity can be determined using the formula for the physical pendulum:

$$g = \omega^2 \cdot l_p \cdot \left(1 + \frac{1}{8} \left(\frac{d_p}{l_p} \right)^2 \right) \quad (18)$$

where the correction term $\frac{1}{8}(d_p/l_p)^2$ takes into account the finite dimensions of the pendulum body (a physical pendulum rather than a mathematical one).

For each of the ten measurement series, a value for T_i (or $\omega_i = 2\pi/T_i$) with the corresponding fit uncertainty $\sigma_{T,i}$ is obtained from the model fit (see Table 3). This yields, for each measurement series, a value g_i with a propagated fit uncertainty $\sigma_{g,i}$ according to Equation 18, where l_p and d_p are initially assumed to be exact, since their error is treated separately (see below).

The final result for g is the average of the ten individual values g_i :

$$\bar{g} = \frac{1}{n} \sum_{i=1}^n g_i = 9.8246 \text{ m/s}^2 \quad (19)$$

Error terms for g . The total error in g consists of three independent components:

σ_{stat} : statistical dispersion of the ten individual measurements g_i around their mean, divided by $\sqrt{n}$ (error in the mean):

$$\sigma_{\text{stat}} = \frac{1}{\sqrt{n}} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (g_i - \bar{g})^2} \quad (20)$$

This component accounts for reproducibility between measurement series (e.g., slight fluctuations in airflow, slightly different starting conditions).

σ_{fit} : Average fit uncertainty from the covariance matrix of the individual model fits:

$$\sigma_{\text{fit}} = \frac{1}{n} \sqrt{\sum_{i=1}^n \sigma_{g,i}^2} \quad (21)$$

This coefficient quantifies how well the model (Eq. 11) describes the respective set of measurements, weighted by the noise σ_U of the Hall probe.

$\sigma_{l,d}$: Error due to the measurement uncertainties of l_p and d_p (previous section), propagated via Equation 18 with the averaged T :

$$\sigma_{l,d} = \left| \frac{\partial g}{\partial l_p} \right| \sigma_{l_p} \oplus \left| \frac{\partial g}{\partial d_p} \right| \sigma_{d_p} \quad (22)$$

Unlike σ_{stat} , this error is systematic: It affects all ten sets of measurements equally (the same length measurement) and therefore does not average out by a factor of $\sqrt{n}$.

The three terms are statistically independent and are added in a quadratic manner:

$$\sigma_g = \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{fit}}^2 + \sigma_{l,d}^2} \quad (23)$$

Contribution	Absolute [m/s ²]	Relative [%]
σ_{stat}	1.04×10^{-3}	0.0105
σ_{fit}	5×10^{-7}	<0.0001
$\sigma_{l,d}$	7.15×10^{-3}	0.0723
σ_g (gesamt)	7.22×10^{-3}	0.0731

Table 4: Error margins for determining g . The relative values are based on $\bar{g}$.

This gives the following final result:

$$g = (9.8246 \pm 0.0072) \text{ m/s}^2 \quad (24)$$

It turns out that the term $\sigma_{l,d}$ from the length measurement of l_p clearly dominates the total error: At 0.0723 %, it accounts for approximately 98 % of the total variance σ_g^2 , while the variation between the ten measurement series (σ_{stat}) contributes only about 2 %, and the pure fitting uncertainty (σ_{fit}) is practically negligible. This means that it is not the electronic measurement of the vibration period, but rather the manual measurement of the length l_1 (using a tape measure, $\sigma \approx 0.494 \text{ mm}$) that limits the accuracy of our result for g . Repeating the experiment with a more precise determination of l_p (e.g., using a vernier caliper instead of a tape measure, or performing multiple independent measurements to reduce σ_{Calib}) would therefore most significantly improve the overall uncertainty on g .

Comparison with the literature value. The gravity value expected by the Bundesamt für Kartographie und Geodäsie (BKG) for the test site is:

$$g_{\text{BKG}} = 9.810836 \text{ m/s}^2. \quad (25)$$

The relative deviation of our result from the reference value is:

$$\frac{\bar{g} - g_{\text{BKG}}}{g_{\text{BKG}}} = 0.142 \%, \quad (26)$$

which corresponds to a deviation of approximately 1.9σ . Thus, our result is still (barely) consistent with the value reported in the literature, although the deviation is primarily attributable to the dominant source of error, $\sigma_{l,d}$.

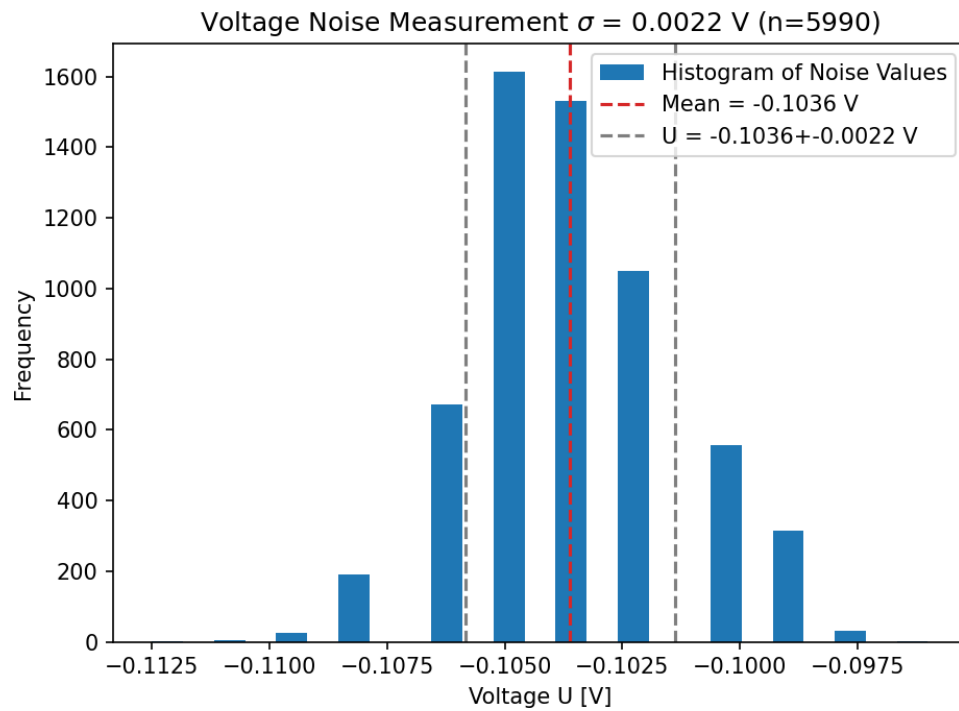
Appendix with additional measurement analyses:

Figure 5: Histogram of the second noise measurement (excluding the first 10 recorded values)

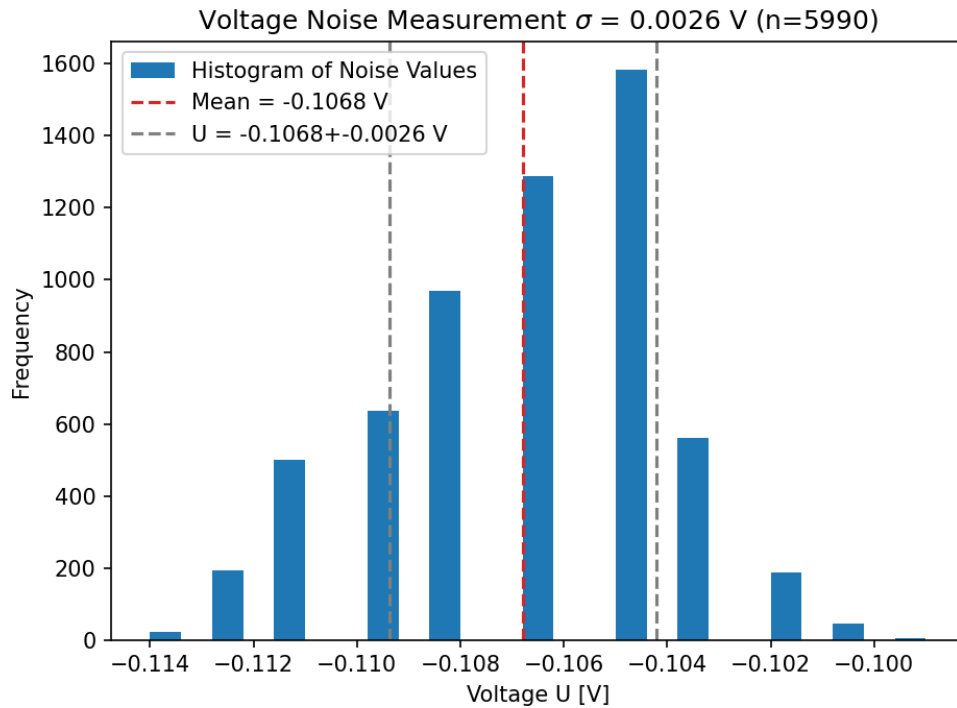


Figure 6: Histogram of the third noise measurement (excluding the first 10 recorded values)

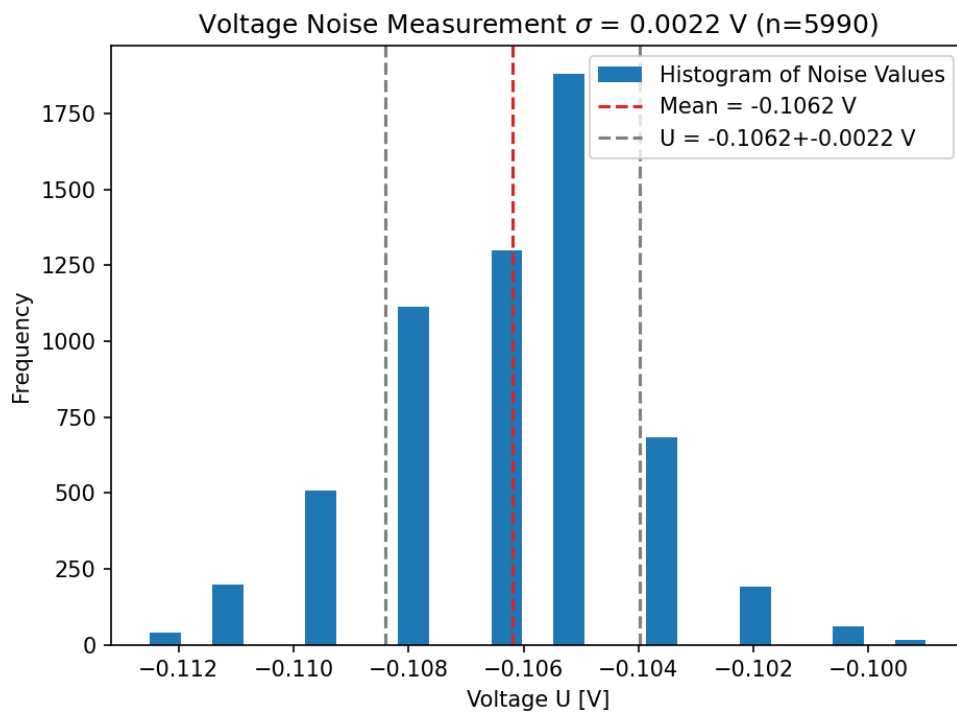


Figure 7: Histogram of the fourth noise measurement (excluding the first 10 recorded values)

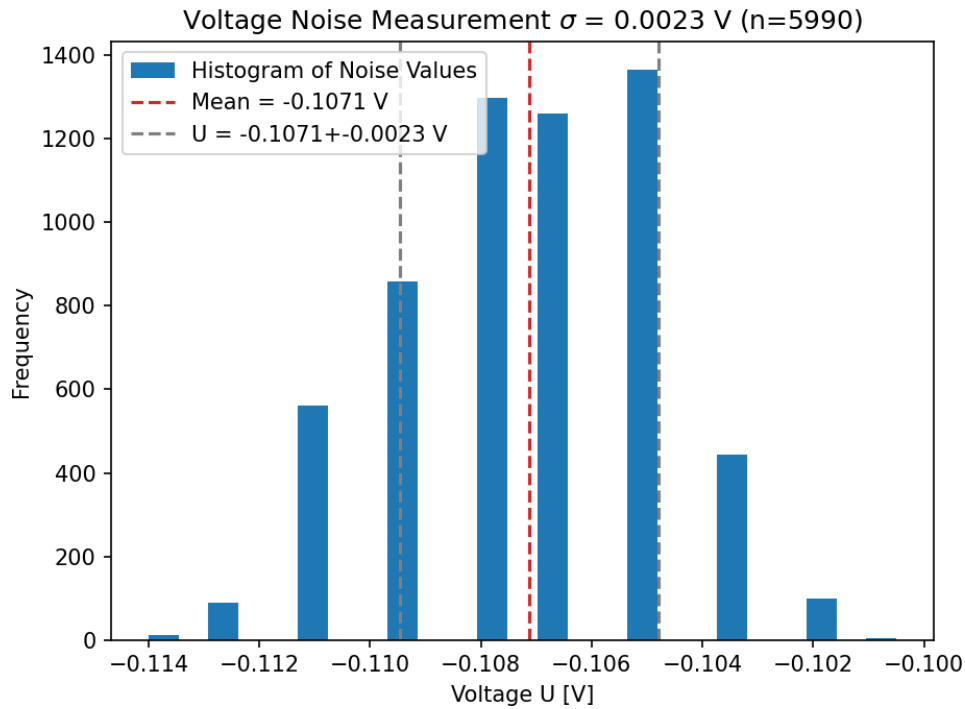


Figure 8: Histogram of the fifth noise measurement (excluding the first 10 recorded values)

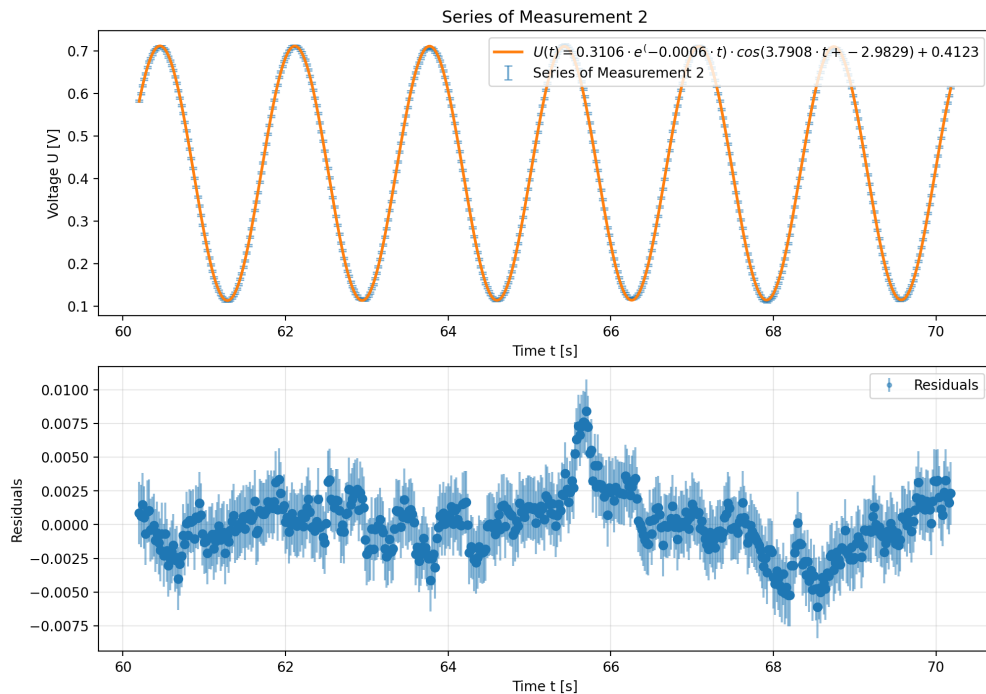


Figure 9: Top: Measurement series 2 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

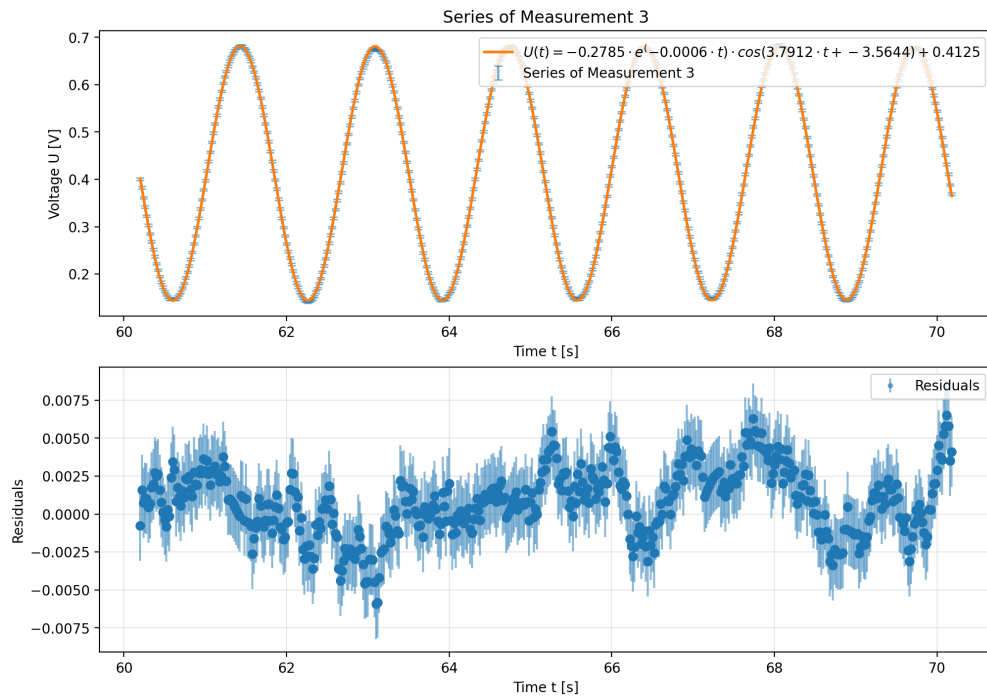


Figure 10: Top: Measurement series 3 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

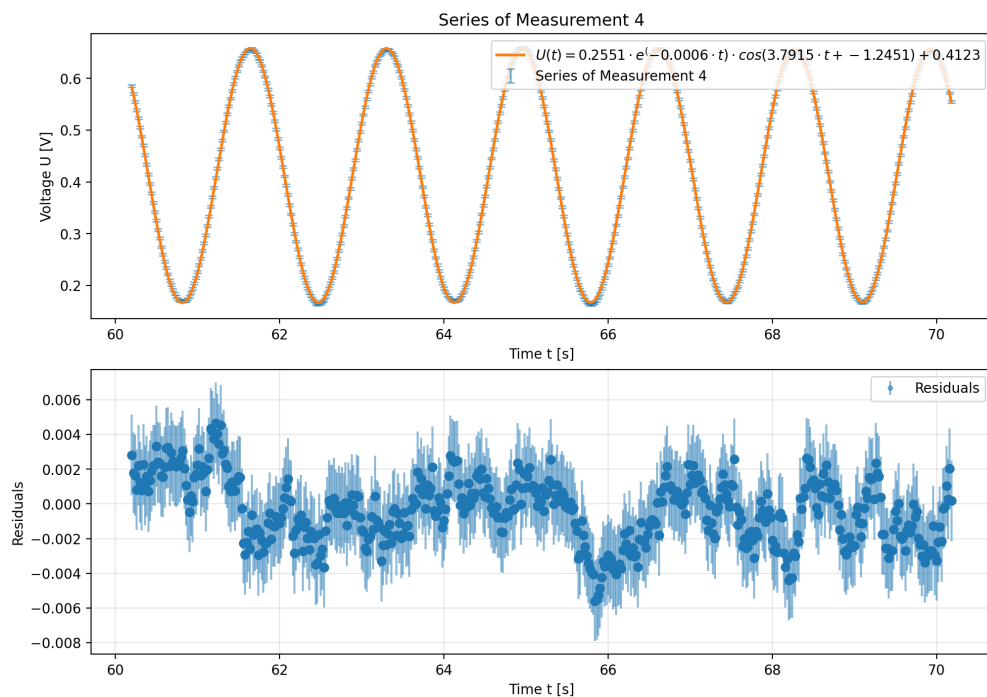


Figure 11: Top: Measurement series 4 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

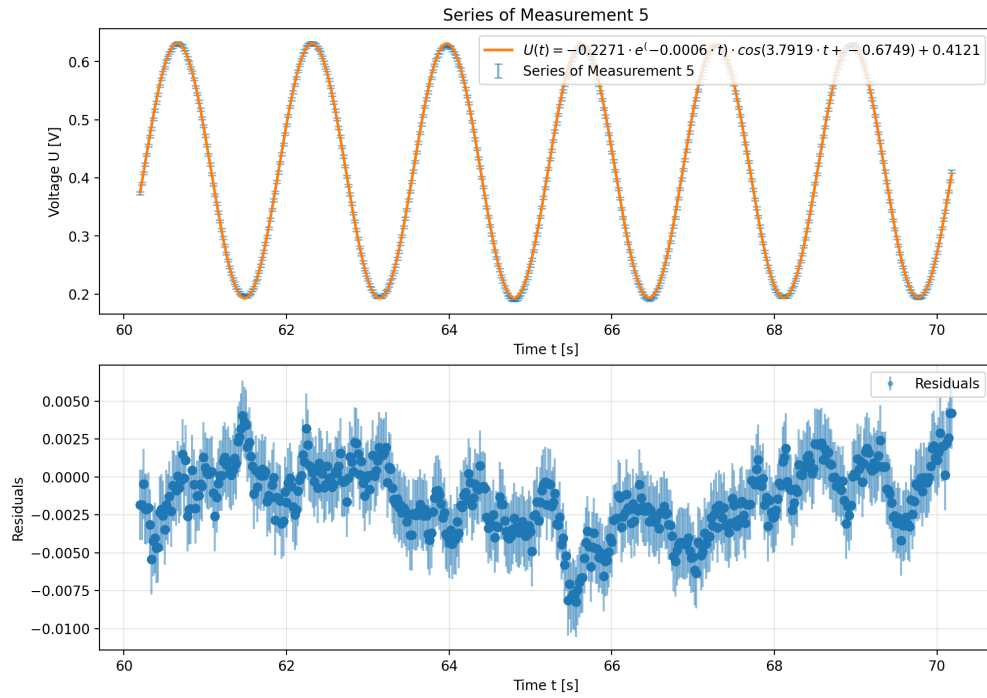


Figure 12: Top: Measurement series 5 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

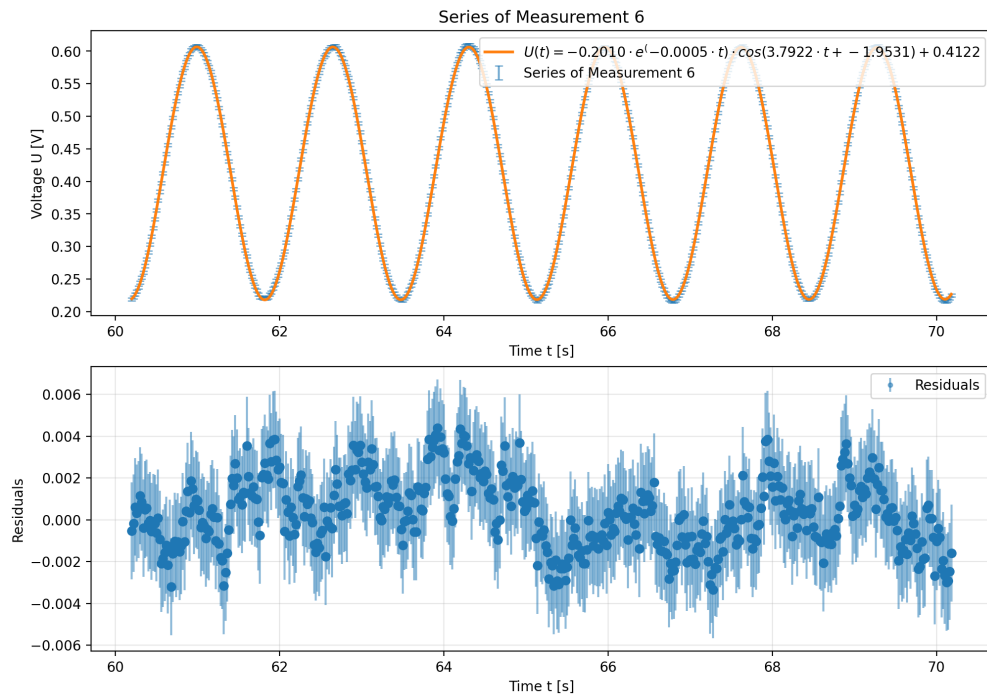


Figure 13: Top: Measurement series 6 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

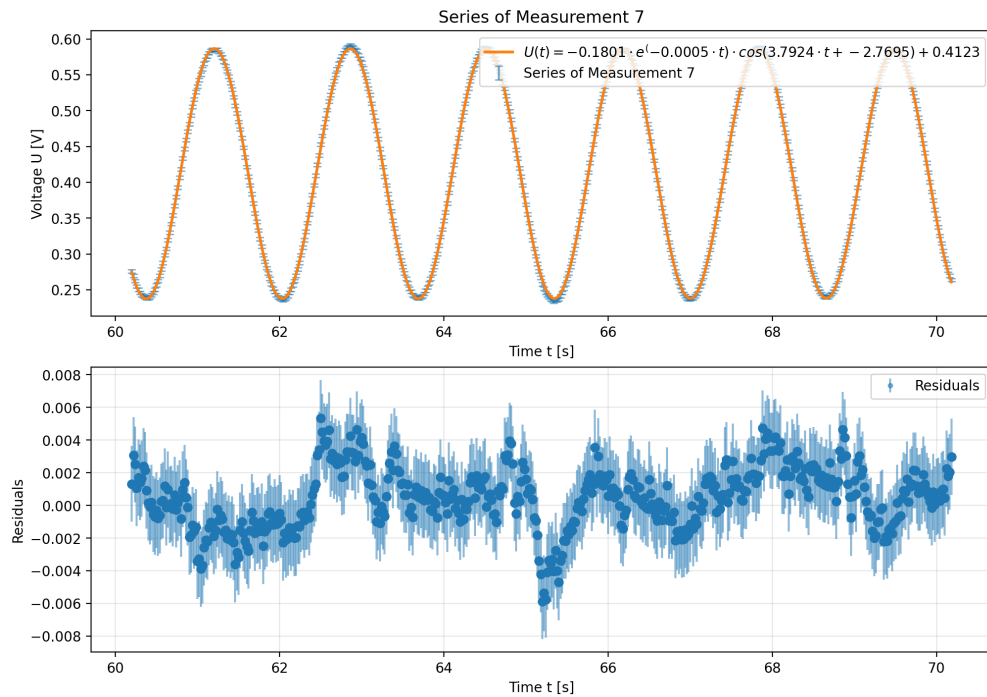


Figure 14: Top: Measurement series 7 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

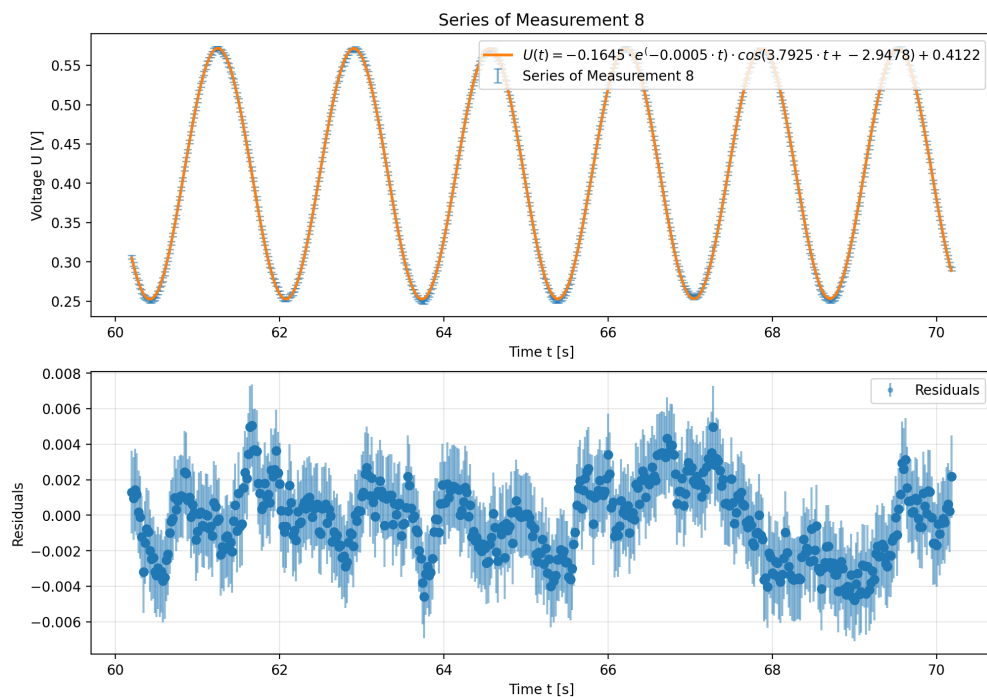


Figure 15: Top: Measurement series 8 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

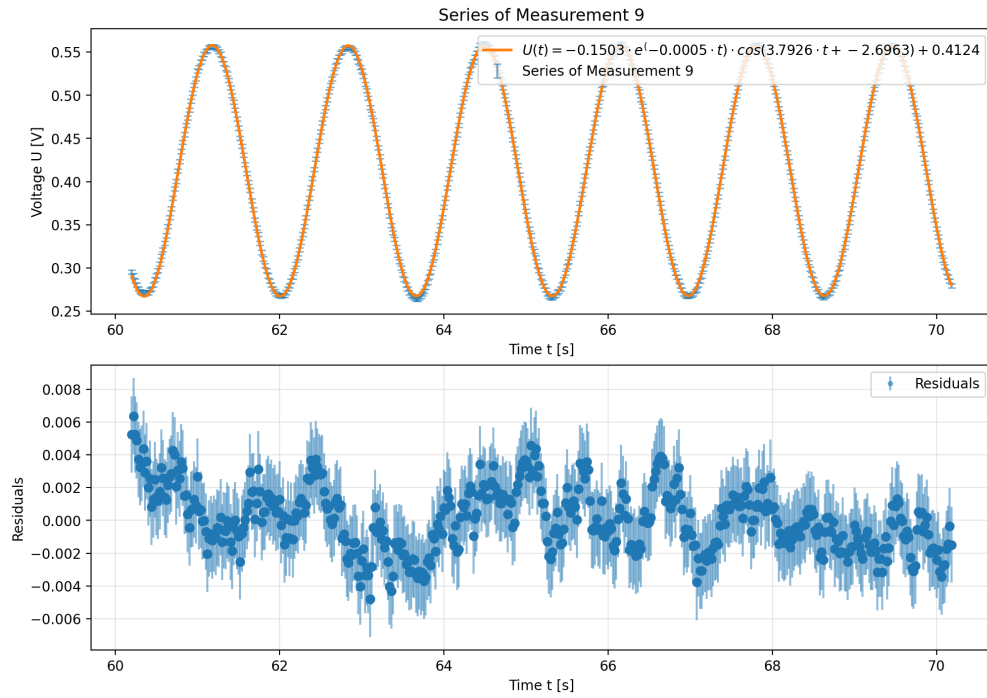


Figure 16: Top: Measurement series 9 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.

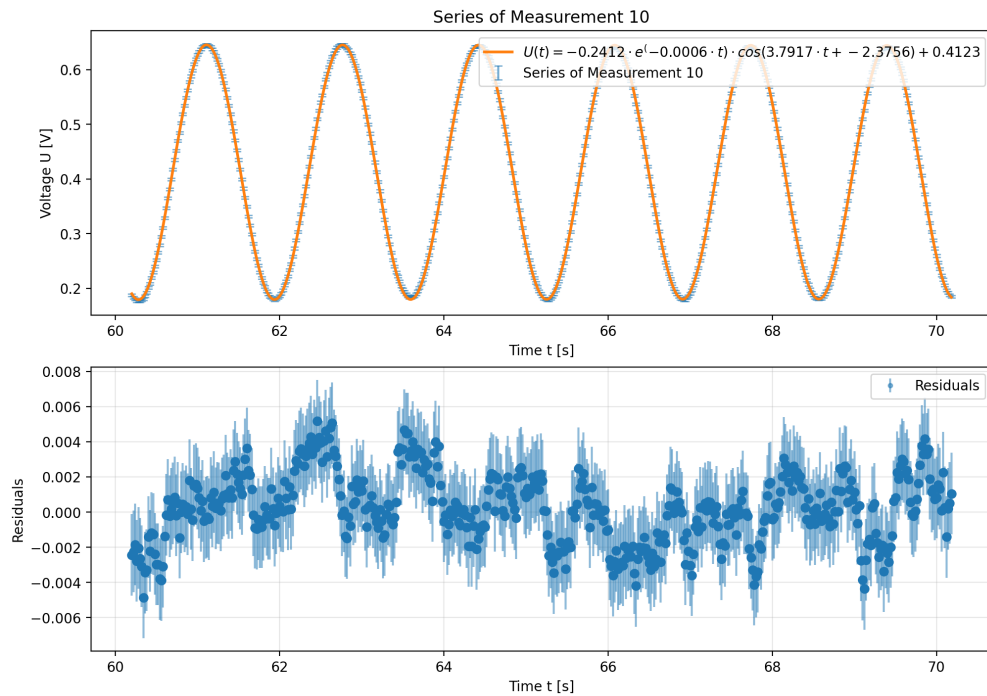


Figure 17: Top: Measurement series 10 (blue, with error bars σ_U) with a fitted model according to Equation 11 (orange). Below: Residuals $U(t) - U_{\text{fit}}(t)$ for the same time interval. The time interval from 60 s to 70 s is shown for clarity.